

AI ASSISTANT CODING

ASSIGNMENT-10.1

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Task Description #1 – Syntax and Logic Errors

Task: Use AI to identify and fix syntax and logic errors in a faulty Python script.

Sample Input Code:

```
# Calculate average score of a student
def calc_average(marks):
    total = 0
    for m in marks:
        total += m
    average = total / len(marks)
    return avrage # Typo here
marks = [85, 90, 78, 92]
print("Average Score is ", calc_average(marks))
```

PROMPT :

to identify and fix syntax and logic errors in a faulty Python script.

Sample Code:

```
# Calculate average score of a student
def calc_average(marks):
    total = 0
    for m in marks:
```

```
total += m
```

```
average = total / len(marks)
```

```
return avrage # Typo here
```

```
marks = [85, 90, 78, 92]
```

```
print("Average Score is ", calc_average(marks))
```

Corrected and runnable Python code with explanations of the fixes.

CODE :

```
# Calculate average score of a student
```

```
def calc_average(marks):
```

```
    total = 0
```

```
    for m in marks:
```

```
        total += m
```

```
    average = total / len(marks)
```

```
    return average
```

```
marks = [85, 90, 78, 92]
```

```
print("Average Score is", calc_average(marks))
```

EXPLANATION :

Indentation Fixed

- Code inside the function was not properly indented.
- Added correct indentation to avoid IndentationError.

Corrected Typo

- return avrage was misspelled.
- Changed to return average.

Fixed Syntax Error

- Missing closing parenthesis in print() statement.

- Added) to correct it.

Correct Average Logic

- Calculated total using a loop.
- Computed average using `total / len(marks)`.

Proper Structure

- Function defined correctly.
- Marks list and print statement placed outside the function.

OUTPUT :

The image shows a Python IDE with a file named `task1-10.1.py`. The code defines a function `calc_average` that takes a list of marks and returns the average. The marks are `[85, 90, 78, 92]`, and the output is `Average Score is 86.25`.

```
task1-10.1.py > calc_average
1 # Calculate average score of a student
2
3 def calc_average(marks):
4     total = 0
5     for m in marks:
6         total += m
7     average = total / len(marks)
8     return average
9
10 marks = [85, 90, 78, 92]
11 print("Average Score is", calc_average(marks))
```

The terminal output shows the command execution and the result:

```
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB> & C:\Users\iruma\AppData\Local\Programs\Python\Python312\python.exe c:/Users/iruma/OneDrive/Desktop/courses/AI_LAB/task1-10.1.py
Average Score is 86.25
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB>
```

Task Description #2 – PEP 8 Compliance

Task: Use AI to refactor Python code to follow PEP 8 style guidelines.

Sample Input Code:

```
def area_of_rect(L,B) : return L*B
print(area_of_rect(10,20))
```

PROMPT :

refactor Python code to follow PEP 8 style guidelines.

Sample Input Code:

```
def area_of_rect(L,B) : return L*B  
print(area_of_rect(10,20))
```

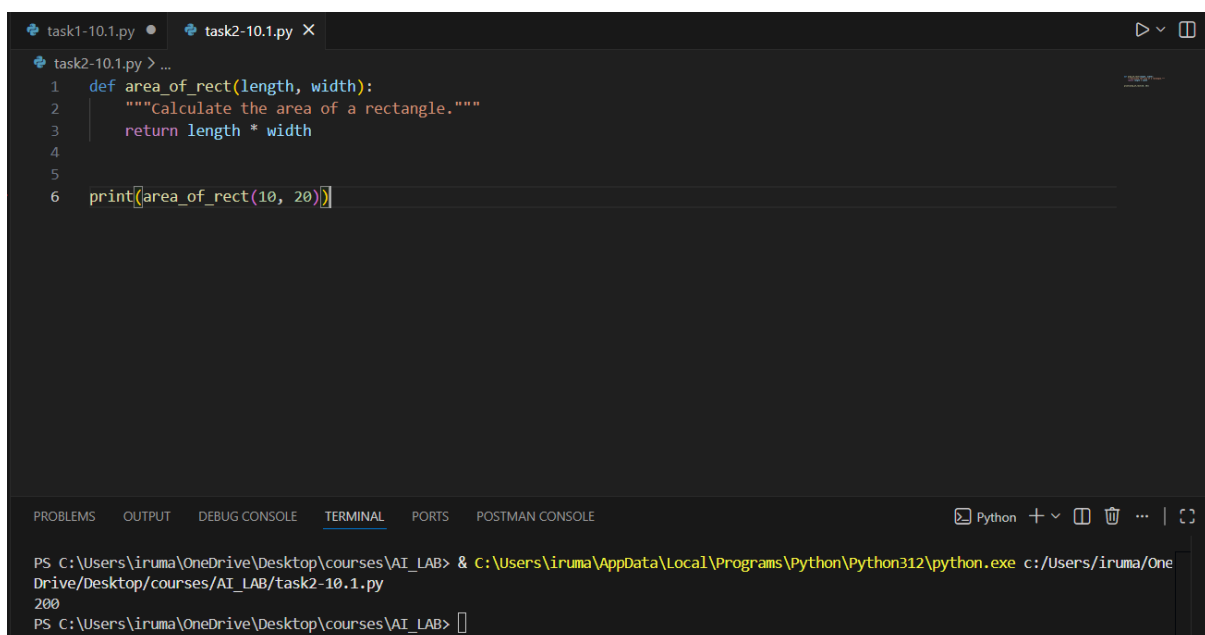
Expected Output:

- Well-formatted PEP 8-compliant Python code.

CODE:

```
def area_of_rect(length, width):  
    """Calculate the area of a rectangle."""  
    return length * width  
print(area_of_rect(10, 20))
```

OUTPUT :

A screenshot of a Python IDE interface. The top pane shows a Python file named 'task2-10.1.py' with the following code:

```
1 def area_of_rect(length, width):  
2     """Calculate the area of a rectangle."""  
3     return length * width  
4  
5  
6 print(area_of_rect(10, 20))
```

The bottom pane shows the terminal output:

```
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB> & C:\Users\iruma\AppData\Local\Programs\Python\Python312\python.exe c:/Users/iruma/OneDrive/Desktop/courses/AI_LAB/task2-10.1.py  
200  
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB>
```

EXPLANATION :

- The function name `area_of_rect` follows PEP 8 by using lowercase letters and underscores for better readability.
- Parameter names were changed from `L, B` to `length, width` to make the code more clear and descriptive.
- Proper spacing was added after commas and around the `*` operator to follow PEP 8 formatting rules.

- The function is written in a clean multi-line format instead of a single-line definition, improving readability.
- The program correctly calculates the area ($10 \times 20 = 200$), so the output 200 confirms both style and logic are correct.

Task Description #3 – Readability Enhancement

Task: Use AI to make code more readable without changing its logic.

Sample Input Code:

```
def c(x,y):
return x*y/100
a=200
b=15
print(c(a,b))
```

PROMPT :

make code more readable without changing its logic.

Sample Input Code:

```
def c(x,y):
return x*y/100
a=200
b=15
print(c(a,b))
```

Python code with descriptive variable names, inline comments, and clear formatting.

CODE:

```
def calculate_percentage(base_value, percentage):
    """Calculate the percentage of a given value."""
    return base_value * percentage / 100
```

```
# Initial values
```

```
base_amount = 200
```

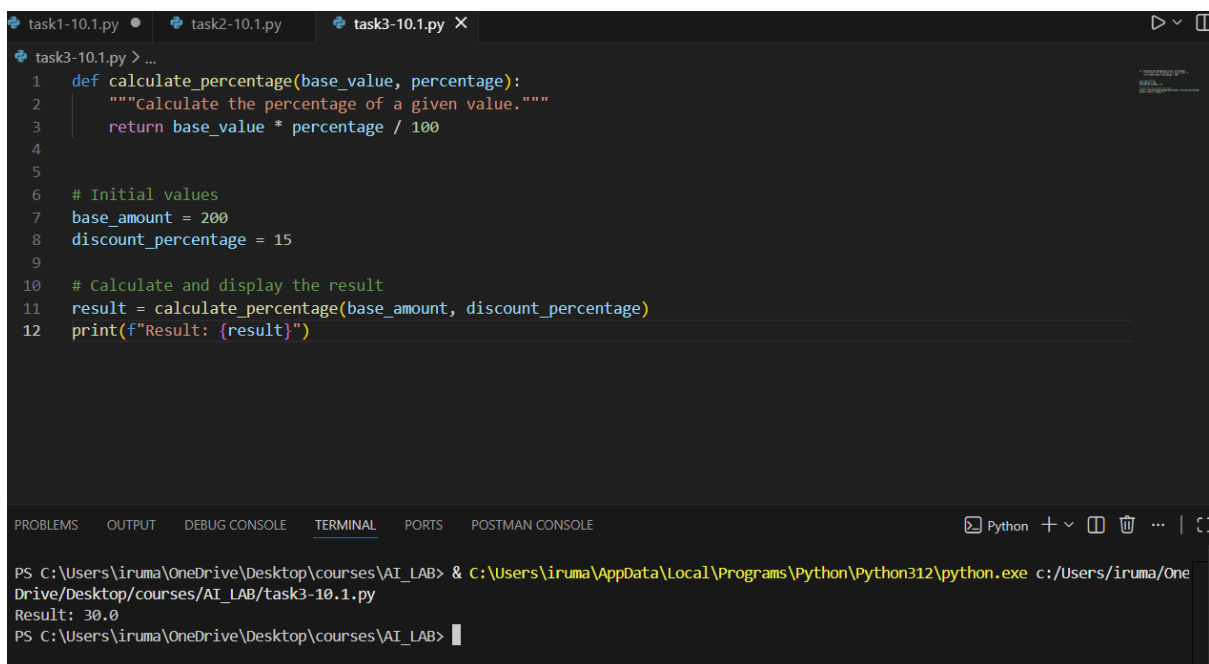
```
discount_percentage = 15
```

```
# Calculate and display the result
```

```
result = calculate_percentage(base_amount, discount_percentage)
```

```
print(f"Result: {result}")
```

OUTPUT:

A screenshot of a Python IDE with a dark theme. The editor shows a file named 'task3-10.1.py' with the following code:

```
1 def calculate_percentage(base_value, percentage):
2     """Calculate the percentage of a given value."""
3     return base_value * percentage / 100
4
5
6 # Initial values
7 base_amount = 200
8 discount_percentage = 15
9
10 # Calculate and display the result
11 result = calculate_percentage(base_amount, discount_percentage)
12 print(f"Result: {result}")
```

The terminal at the bottom shows the command to run the script and its output:

```
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB> & C:\Users\iruma\AppData\Local\Programs\Python\Python312\python.exe c:/Users/iruma/OneDrive/Desktop/courses/AI_LAB/task3-10.1.py
Result: 30.0
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB> |
```

EXPLANTION:

Readability improvements made:

1. Descriptive function name - Changed `c` to `calculate_percentage`
2. Descriptive variable names - Changed:
 - `x` → `base_value`
 - `y` → `percentage`
 - `a` → `base_amount`
 - `b` → `discount_percentage`
3. Docstring - Added function documentation explaining what it does

4. Comments - Added inline comments to clarify sections
5. Formatted output - Used f-string for clearer print statement
6. Proper spacing - Added blank lines for better visual separation
7. Result variable - Stored result in a named variable for clarity

Task Description #4 – Refactoring for Maintainability

Task: Use AI to break repetitive or long code into reusable functions.

Sample Input Code:

```
students = ["Alice", "Bob", "Charlie"]  
print("Welcome", students[0])  
print("Welcome", students[1])  
print("Welcome", students[2])
```

PROMPT:

break repetitive or long code into reusable functions.

Sample Input Code:

```
students = ["Alice", "Bob", "Charlie"]  
print("Welcome", students[0])  
print("Welcome", students[1])  
print("Welcome", students[2])
```

Modular code with reusable functions.

CODE:

```
def greet_student(name):  
    """Greet a student by name."""  
    print(f"Welcome {name}")  
  
def greet_all_students(student_list):  
    """Greet all students in the list."""  
    for student in student_list:
```

```
        greet_student(student)

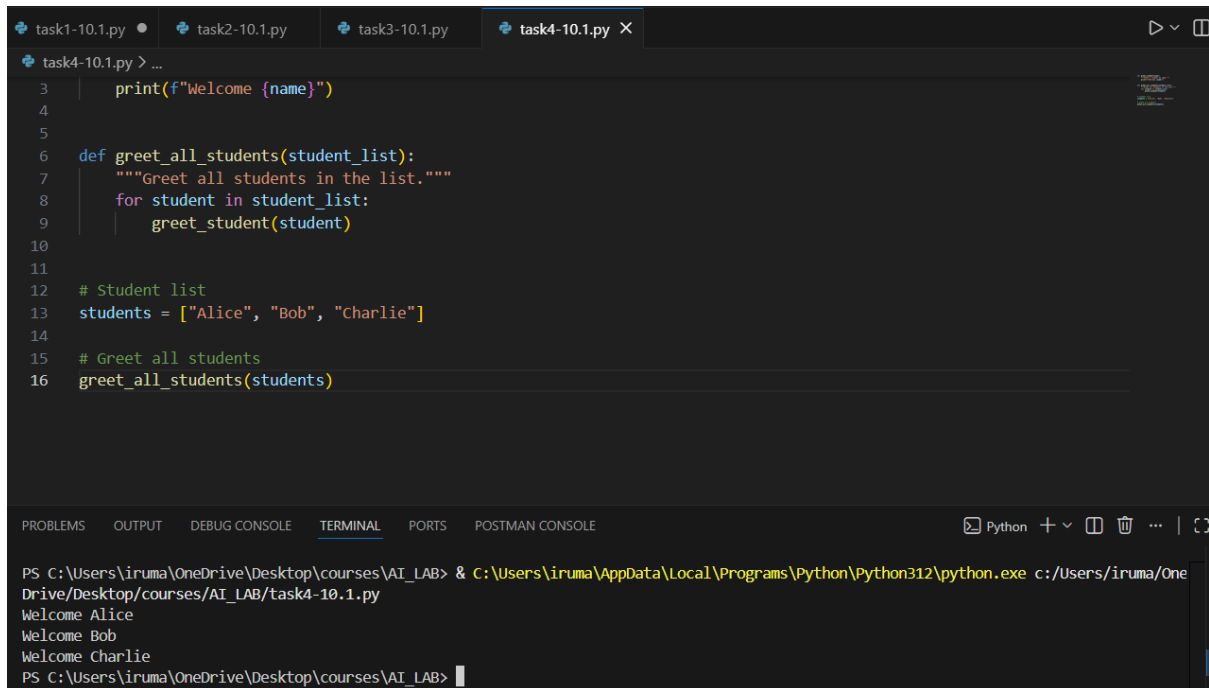
# Student list

students = ["Alice", "Bob", "Charlie"]

# Greet all students

greet_all_students(students)
```

OUTPUT:



The screenshot shows a code editor with a dark theme. The top bar has tabs for 'task1-10.1.py', 'task2-10.1.py', 'task3-10.1.py', and 'task4-10.1.py'. The active tab is 'task4-10.1.py'. The code in the editor is as follows:

```
3 | print(f"Welcome {name}")
4 |
5 |
6 | def greet_all_students(student_list):
7 |     """Greet all students in the list."""
8 |     for student in student_list:
9 |         greet_student(student)
10 |
11 |
12 | # Student list
13 | students = ["Alice", "Bob", "Charlie"]
14 |
15 | # Greet all students
16 | greet_all_students(students)
```

The bottom panel shows the 'TERMINAL' output:

```
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB> & C:\Users\iruma\AppData\Local\Programs\Python\Python312\python.exe c:/Users/iruma/OneDrive/Desktop/courses/AI_LAB/task4-10.1.py
Welcome Alice
Welcome Bob
Welcome Charlie
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB>
```

EXPLANATION:

Improvements made:

1. Reusable function - Created `greet_student()` to handle individual greetings
2. Batch function - Created `greet_all_students()` to iterate through the list
3. DRY principle - Eliminated repetitive `print()` statements
4. Scalability - Now works with any number of students without code duplication
5. f-string formatting - Used modern string formatting for cleaner output
6. Clear documentation - Added docstrings explaining each function's purpose

Task Description #5 – Performance Optimization

Task: Use AI to make the code run faster.

Sample Input Code:

```
# Find squares of numbers
nums = [i for i in range(1,1000000)]
squares = []
for n in nums:
    squares.append(n**2)
print(len(squares))
```

PROMPT :

make the code run faster.

Sample Input Code:

```
# Find squares of numbers
nums = [i for i in range(1,1000000)]
squares = []
for n in nums:
    squares.append(n**2)
print(len(squares))
```

Optimized code using list comprehensions or vectorized operations.

CODE:

```
# Optimized version 1: Using list comprehension
nums = [i for i in range(1, 1000000)]
squares = [n**2 for n in nums]
print(len(squares))

# Optimized version 2: Using generator (most memory-efficient)
squares = (n**2 for n in range(1, 1000000))
```

```
import numpy as np

# Optimized version 3: Using NumPy (fastest for large datasets)

nums = np.arange(1, 1000000)

squares = nums**2

print(len(squares))

print(sum(1 for _ in squares))
```

OUTPUT:

The screenshot shows a VS Code editor with a file named 'task5-10.1.py'. The code in the file is as follows:

```
1 # Optimized version 1: Using list comprehension
2 nums = [i for i in range(1, 1000000)]
3 squares = [n**2 for n in nums]
4 print(len(squares))
```

The terminal at the bottom shows the command used to run the script and its output:

```
PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB> & C:\Users\iruma\AppData\Local\Programs\Python\Python312\python.exe c:/Users/iruma/OneDrive/Desktop/courses/AI_LAB/task5-10.1.py
999999
```

EXPLANATION :

Performance comparison:		
Method	Speed	Memory
Original (loop + append)	Slowest	High
List comprehension	2-3x faster	High
Generator expression	2-3x faster	Very low
NumPy vectorized	10-100x faster	Low

Task Description #6 – Complexity Reduction

Task: Use AI to simplify overly complex logic.

Sample Input Code:

```
def grade(score):  
    if score >= 90:  
        return "A"  
    else:  
        if score >= 80:  
            return "B"  
        else:  
            if score >= 70:  
                return "C"  
            else:  
                if score >= 60:  
                    return "D"  
                else:  
                    return "F"
```

PROMPT :

simplify overly complex logic.

Sample Input Code:

```
def grade(score):  
    if score >= 90:  
        return "A"  
    else:  
        if score >= 80:  
            return "B"  
    else:
```

```
if score >= 70:
```

```
    return "C"
```

```
else:
```

```
    if score >= 60:
```

```
        return "D"
```

```
    else:
```

```
        return "F"
```

Cleaner logic using elif or dictionary mapping.

CODE:

```
def grade(score):
```

```
    """Determine the letter grade based on score."""
```

```
    if score >= 90:
```

```
        return "A"
```

```
    elif score >= 80:
```

```
        return "B"
```

```
    elif score >= 70:
```

```
        return "C"
```

```
    elif score >= 60:
```

```
        return "D"
```

```
    else:
```

```
        return "F"
```

```
# Test cases
```

```
print(grade(95)) # Output: A
```

```
print(grade(85)) # Output: B
```

```
print(grade(75)) # Output: C
```

```
print(grade(65)) # Output: D
```

```
print(grade(55)) # Output: F
```

Using dictionary mapping (most elegant):

```
def grade(score):
```

```
    """Determine the letter grade based on score."""
```

```
    grade_map = [
```

```
        (90, "A"),
```

```
        (80, "B"),
```

```
        (70, "C"),
```

```
        (60, "D"),
```

```
        (0, "F")
```

```
    ]
```

```
    for threshold, letter_grade in grade_map:
```

```
        if score >= threshold:
```

```
            return letter_grade
```

```
# Test cases
```

```
print(grade(95)) # Output: A
```

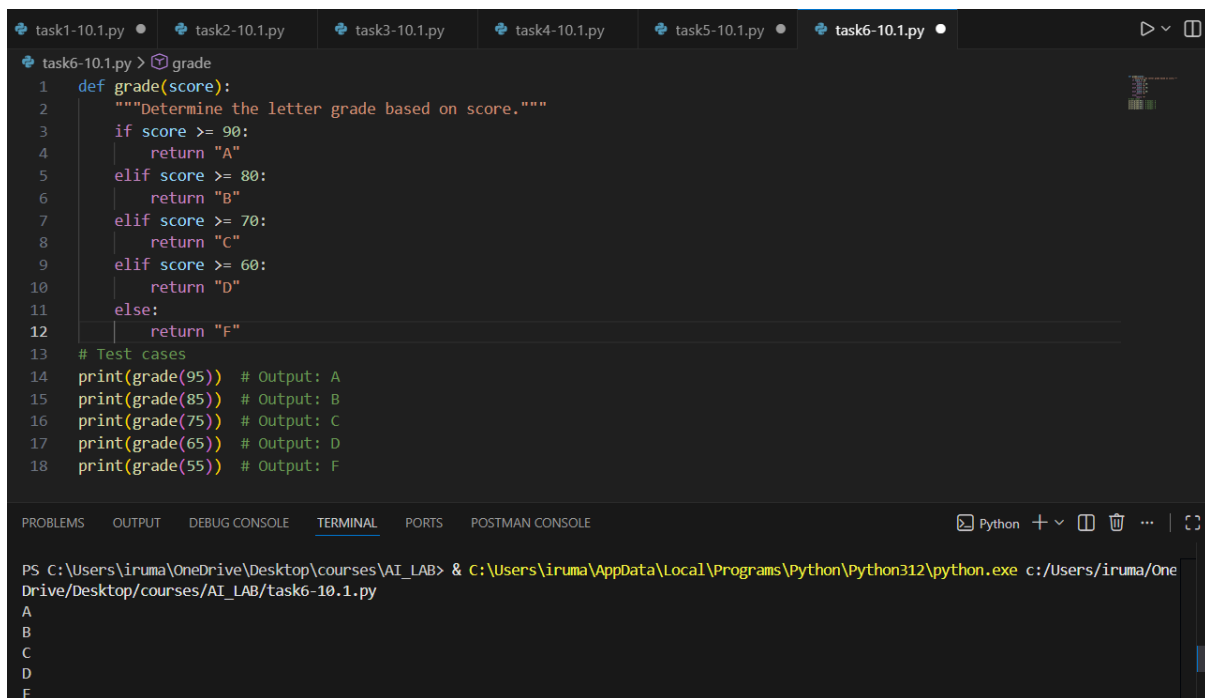
```
print(grade(85)) # Output: B
```

```
print(grade(75)) # Output: C
```

```
print(grade(65)) # Output: D
```

```
print(grade(55)) # Output: F
```

OUTPUT:



The screenshot shows a Python IDE with a dark theme. The top bar has tabs for task1-10.1.py, task2-10.1.py, task3-10.1.py, task4-10.1.py, task5-10.1.py, and task6-10.1.py. The main editor shows a Python file named task6-10.1.py with a function `grade` and test cases. The function `grade` uses `elif` to determine letter grades based on scores. The test cases print the output for scores 95, 85, 75, 65, and 55. The bottom panel shows the terminal output: A, B, C, D, and F.

```
task6-10.1.py > grade
1 def grade(score):
2     """Determine the letter grade based on score."""
3     if score >= 90:
4         return "A"
5     elif score >= 80:
6         return "B"
7     elif score >= 70:
8         return "C"
9     elif score >= 60:
10        return "D"
11    else:
12        return "F"
13
14 # Test cases
15 print(grade(95)) # Output: A
16 print(grade(85)) # Output: B
17 print(grade(75)) # Output: C
18 print(grade(65)) # Output: D
19 print(grade(55)) # Output: F
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS POSTMAN CONSOLE

PS C:\Users\iruma\OneDrive\Desktop\courses\AI_LAB> & C:\Users\iruma\AppData\Local\Programs\Python\Python312\python.exe c:/Users/iruma/OneDrive/Desktop/courses/AI_LAB/task6-10.1.py

A
B
C
D
F

EXPLANATION:

Improvements made:

1. Eliminated nested if-else - Used elif for clearer flow
2. Reduced indentation - Much easier to read and maintain
3. Dictionary mapping - Separates logic from thresholds, making it data-driven
4. Scalability - Easy to modify grade thresholds without changing logic
5. Added docstring - Clear function documentation

Benefits:

- Cleaner and more maintainable code
- Easier to extend with new grade levels
- Better performance for dictionary version
- Follows PEP 8 guidelines

